

Pension Run

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1. Introduction

TBD.

Related Literature:

- [Bi and Zubairy \(2023\)](#). Empirical paper on the mechanisms of pension reforms. Three mechanisms are important: income effect on working longer caused by benefit cut, early withdrawal to extract more benefits, and the uncertainty mechanism due to implementation uncertainty.

2. Model

We consider a continuous time model with an infinite horizon. There is a continuum of individuals choosing between the lumpsum and the monthly pension schemes. The pension fund is running by the government with the return and the volatility being exogenously given. When the pension system goes bankrupt, the recipients cannot receive any more pensions from then on.

2.1. Population

Let $b(s), m(s)$ be the fertility rate and the mortality rate respectively. Let $g(t, s)$ be the unnormalized population density. The population dynamic is given by

$$\frac{\partial g}{\partial t} = - \underbrace{\frac{\partial g}{\partial s}}_{\text{aging}} - \underbrace{mg}_{\text{mortality}} + \underbrace{\int bgds\delta_0}_{\text{birth}} \quad (1)$$

where δ_0 denotes the Dirac delta centered at 0. To simplify the model, we take the assumptions that $b(s) = b$ is a constant and that $m(s) = m\mathbf{1}\{s \geq T_m\}$. In plain words, this means that only those who have aged T_m exposes to the mortality risks.¹

¹We recognize that this is a somewhat restrictive and unrealistic assumption, but as we are going to see, this assumption helps to greatly simplify the computation.

Let $N_t = \int_0^\infty g(t, s) ds$ be the total population and $\hat{g}(t, s) = \frac{g(t, s)}{N_t}$ be the normalized population density. Assuming that $\hat{g}$ is at its steady state, an analytical solution is available:

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m \\ be^{-ns} e^{-m(s-T_m)} & s \geq T_m \end{cases} \quad (2)$$

where $n = \frac{\dot{N}_t}{N_t}$ is the population growth rate satisfying

$$1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}. \quad (3)$$

For a detailed derivation, we refer to [appendix A](#).

2.2. Pension System

We assume that all households retire at age T . When retiring, the households choose between a lumpsum transfer ℓ and a monthly pension that pays a stream of income p . Let $Q(t, s)$ be the population density among the monthly pension recipients. The dynamic is given by

$$\frac{\partial Q}{\partial t} = - \underbrace{\frac{\partial Q}{\partial s}}_{\text{aging}} - \underbrace{mQ}_{\text{mortality}} + \underbrace{q_t \hat{g}(T) N_t \delta_T}_{\text{new retirees}} \quad (4)$$

where q_t is the proportion of new retirees choosing pension. Suppose that $T \geq T_m$. We can further simplify the recipients dynamic thanks to our assumption on the mortality rate. Let $Q_t = \int_T^\infty Q(t, s) ds$. Q_t follows the law of motion

$$dQ_t = [q_t \hat{g}(T) N_t - mQ_t] dt. \quad (5)$$

Let B_t be the level of the pension fund. Its law of motion is given by

$$dB_t = \left[\underbrace{rB_t}_{\text{interest}} + \underbrace{\tau y N_t \int_{T_w}^T \hat{g}(s) ds}_{\text{pension tax}} - \underbrace{(1 - q_t) \ell N_t \hat{g}(T)}_{\text{lumpsum}} - \underbrace{pQ_t}_{\text{monthly}} \right] dt + \sigma B_t dZ_t, \quad (6)$$

where Z_t is a Brownian motion. τ is the tax rate; y is the income ; T_w is the age when individuals start to work.

The system goes bankrupt when B_t hits 0. Denote the remaining time for B_t to hit 0 by T_B . We shall consider the detrended states $\hat{B}_t = \frac{B_t}{N_t}$, $\hat{Q}_t = \frac{Q_t}{N_t}$. Since N_t is positive in finite time, the bankrupt time is equivalent to the time when $\hat{B}_t = 0$. In other words, the the remaining time for $\hat{B}_t$ hitting 0, $T_{\hat{B}}$, coincides with T_B . The law of motions for the transformed states

are given by

$$d\hat{B}_t = \left[(r - n)\hat{B}_t + \tau y \int_{T_w}^T \hat{g}(s) ds - (1 - q_t)\hat{g}(T)\ell - p\hat{Q}_t \right] dt + \sigma \hat{B}_t dZ_t \quad (7)$$

$$d\hat{Q}_t = [q_t \hat{g}(T) - (m + n)\hat{Q}_t] dt. \quad (8)$$

To simplify the notation, we also denote the drift terms by μ_B and μ_Q , respectively.

2.3. Individuals' Pension Decisions

A retiring individual i chooses between the two schemes $d \in \{\ell, p\}$ to maximize the utility, given by

$$u_{it}(d) = \begin{cases} \alpha + \ell + \beta \epsilon_i(\ell) & \text{if } d = \ell, \\ V_t + \beta \epsilon_i(p) & \text{if } d = p, \end{cases} \quad (9)$$

where V_t is the present discount value of the monthly pension given the current states $(\hat{B}_t, \hat{Q}_t)$,

$$V_t = V(\hat{B}_t, \hat{Q}_t) = \mathbb{E} \left[\int_0^{T_B} e^{-(\rho+m)s} p ds \mid \hat{B}_t, \hat{Q}_t \right] \quad (10)$$

with ρ being the subjective discount factor. We assume that the $\epsilon_i(d)$ are independent and identically distributed across individuals and the choices, following the standard Gumbel distribution. The shift parameter α controls the preference of the lumpsum scheme over the monthly one. Although we do not model here elaborately, this may be thought to represent the need of liquidity or the incomplete pension market. Another parameter β controls the sensitivity to the fluctuation in the present discount value of the monthly pension.

By the property of the Gumbel distribution, the individuals' choices aggregate to the proportion choosing the monthly scheme

$$q_t = P_t(u_{it}(p) > u_{it}(\ell)) = \frac{1}{1 + \exp\left(\frac{\alpha + \ell - V_t}{\beta}\right)} \quad (11)$$

by the law of large number across the individuals. Note that q_t is a function of $\hat{B}_t$ and $\hat{Q}_t$ since $V_t = V(\hat{B}_t, \hat{Q}_t)$.

2.4. Equilibrium

We now list down the equilibrium conditions. Given the Markovian states $(\hat{B}_t, \hat{Q}_t)$, the equilibrium is determined by the present discount value of the monthly pension V_t and the proportion choosing monthly scheme q_t such that

- (a) Given q_t , the law of motions of the states $(\hat{B}_t, \hat{Q}_t)$ is consistent with q_t (equation (7)).
- (b) Given the law of motions of the states, the present discount value of the monthly pension $V(\hat{B}_t, \hat{Q}_t)$ is consistent with the law of motions (equation (10)).

(c) The individuals optimize and the decisions aggregate to q_t (equation (11)).

To determine the equilibrium, it suffices to compute $V(\hat{B}, \hat{Q})$. Notice that V satisfies the following Hamilton-Jacobi-Bellman equation

$$(\rho + m)V = p + \mathcal{A}V, \quad (12)$$

with the boundary condition $V(0, \hat{Q}) = 0$ for all $\hat{Q}$. The operator

$$\mathcal{A} = \mu_B \frac{\partial}{\partial B} + \mu_Q \frac{\partial}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2}{\partial B^2} \quad (13)$$

is the infinitesimal generator. Since the drift terms also depend on V , this partial differential equation is nonlinear.

To solve the model numerically, we impose the artificial boundary at the upper bound of $\hat{B}$. Since clearly V_t has upper bound $\frac{p}{\rho+m}$ and is increasing in $\hat{B}$, it follows that $\frac{\partial V}{\partial B}$ is approximately 0 for $\hat{B}$ sufficiently large.

We solve the model based on the finite difference method alongside the successive iteration that shares the same spirit with Howard's policy iteration method. This numerical scheme can be viewed as an semi-implicit scheme with time step size being infinite. The sparsity of the generator matrix is exploited to achieve a better efficiency. We present our algorithm explicitly in algorithm 1.

3. Results

3.1. Calibration

We separate the parameters in the model in to two parts. One is calibrated outside the model and the other is estimated by solving the inverse problem. To illustrate the estimation procedure, denote the estimand vector θ . The data is a time series $X_i = (\hat{B}_i, \hat{Q}_i)$ sampled from equidistant time grids of gap size Δ . The inverse problem is the following:

$$\min_{\theta} Q(f, \theta) \quad (14)$$

subject to

$$((\rho + m)I - \mathcal{A}(V, \theta))V = p \quad (15)$$

$$e^{\Delta(\mathcal{A}(V, \theta))^*} \delta_x = f(\cdot, x) \quad (16)$$

where Q is some type of the objective function of the M-estimator. The first constraint is the backward equation and the second one is the forward equation. The Lagrangian is given by

$$L(\theta, \lambda_b, \lambda_f) = Q(f, \theta) + \langle \lambda_b, ((\rho + m)I - \mathcal{A}(V, \theta))V - p \rangle + \langle \lambda_f, e^{\Delta(\mathcal{A}(V, \theta))^*} \delta_x - f \rangle. \quad (17)$$

Algorithm 1 Successive Iteration

Require: Initial guess $V^{(0)}$, tolerance $\epsilon > 0$, maximum iterations K

1: Set $k \leftarrow 0$

2: **repeat**

3: Given $V^{(k)}$, compute

$$\mu_B^{(k)} = \mu_B(B, Q, V^{(k)}(B, Q)), \quad \mu_Q^{(k)} = \mu_Q(B, Q, V^{(k)}(B, Q))$$

and the infinitesimal generator

$$\mathcal{A}^{(k)} = \mu_B^{(k)} \frac{\partial}{\partial B} + \mu_Q^{(k)} \frac{\partial}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2}{\partial B^2}.$$

with adjustments to the boundary nodes.

4: Solve the linear system

$$\left((\rho + m)I - \mathcal{A}^{(k)} \right) V^{(k+1)} = \mathbf{b}, \quad \mathbf{b} = p \operatorname{vec} \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & 1 \end{pmatrix}.$$

5: Compute the error

$$e^{(k+1)} \leftarrow \left\| V^{(k+1)} - V^{(k)} \right\|_{\infty}.$$

6: $k \leftarrow k + 1$

7: **until** $e^{(k)} < \epsilon$ or $k = K$

8: **return** $V^{(k)}$.

Differentiate with respect to θ .

$$\frac{\partial L}{\partial \theta} = \frac{\partial Q}{\partial \theta} + \langle \lambda_b, \partial_{\theta} [((\rho + m)I - \mathcal{A}(V, \theta))V] \rangle + \langle \lambda_f, \partial_{\theta} e^{\Delta(\mathcal{A}(V, \theta))^*} \delta_x \rangle$$

The parameters for the population structure are calibrated to Taiwan's population. We take $T_m = T_r = 65$ and $T_w = 20$. The birth rate is set to be the total fertility rate in 2025 and the mortality rate leads to the median lifespan being 77. The details are given in [table 1](#).

Appendix

A. Derivation of the Population Equations

To find the steady state $\hat{g}$, conjecture that $g(t, s) = \hat{g}(s)N_t$. Plugging back to the population equation gives

$$\hat{g}(s) \frac{\dot{N}_t}{N_t} = -\frac{\partial \hat{g}}{\partial s} - m(s)\hat{g}(s) + b\delta_0. \quad (18)$$

Table 1: Model Parameters

Parameter	Description	Value	Target
<i>Panel A: Parameters Calibrated Outside the Model</i>			
b	Birth rate	0.00462	Total fertility rate
m	Mortality rate	0.058	Median lifespan
T_w	Working age	20	
T_r	Retirement age	65	Standard retirement age
T_m	Age of mortality exposure	65	
ρ	Subjective discount rate	0.02	Interest rate
τ	Pension tax	0.125	Pension tax
y	Income	5.496	Highest Monthly Insured Salry
ℓ	Lumpsum pension	20.61	Transfer based on 30 years of working
p	Monthly Pension	2.556	Pension based on 30 years of working
<i>Panel B: Parameters Estimated Within the Model</i>			
r	Pension fund return rate	0.041	
σ	Pension fund volatility	0.083	
α	Preference on lumpsum scheme	-7.752	
β	Sensitivity to default risk	2.623	

Note: The monetary unit in the model is 100,000 NTD.

Notice that the right hand side does not depend on t . Hence $\frac{\dot{N}_t}{N_t} = n$ is a constant. We can also write the Dirac delta as part of the boundary condition:

$$\hat{g}(s)n = -\frac{\partial \hat{g}}{\partial s} - m(s)\hat{g}(s), \quad \hat{g}(0) = b. \quad (19)$$

This is a standard initial value problem with $m(s) = m$ for $s \geq T_m$ and 0 otherwise. The solution is given by

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m \\ be^{-ns}e^{-m(s-T_m)} & s \geq T_m. \end{cases} \quad (20)$$

Finally, the population growth rate n can be determined by the normalizing condition

$$\int_0^\infty \hat{g}(s)ds = 1 \quad \Rightarrow \quad 1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}. \quad (21)$$

References

Bi, H., & Zubairy, S. (2023, November). Public pension reforms and retirement decisions: Narrative evidence and aggregate implications. *American Economic Journal: Economic Policy*, 15(4), 142–82.